

FOCUS HEALTH

Exploring the growing role of exercise in improving quality of life and outcomes in lung cancer care

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Lung cancer continues to be one of the most prevalent and deadliest cancers globally. With more than two million new cases diagnosed each year, its impact extends beyond physical health, affecting patients emotionally, psychologically, and socially. Recent research highlights an often-overlooked yet powerful ally in the fight against lung cancer: **exercise.**

Exercise can play a major role in lung cancer management—from diagnosis through to advanced disease stages. The evidence suggests that when prescribed and supervised appropriately, exercise is not only safe but beneficial for patients at all stages of treatment.

Why Exercise Matters in Lung Cancer Care

Lung cancer and its treatments-including surgery, chemotherapy, radiation, and targeted therapies-often result in debilitating side effects like shortness of breath (dyspnoea), fatigue, anxiety, and loss of muscle mass. These effects can severely diminish a patient's ability to carry out daily activities and may even deter them from adhering to their treatment plans.

According to the research studies, structured exercise interventions-especially those combining **cardiorespiratory training** and **resistance exercise**-can lead to improvements in lung function, muscle strength, mood, and overall quality of life. Notably, patients who engage in regular physical activity before and after diagnosis have shown improved survival rates and better tolerance to cancer treatments.

The Evidence for Exercise

The authors evaluated a broad range of studies and found that exercise delivered across various formats and intensities had significant positive outcomes. For example:

- **Preoperative exercise** improves cardiorespiratory fitness, reduces anxiety, and lowers the risk of post-surgical complications.
- **Postoperative exercise** helps restore physical function, reduces hospital stays, and improves mental well-being.
- **Exercise during treatment** has been linked to reduced fatigue, better cardiovascular health, and fewer treatment interruptions.
- Even in **advanced stages** or in patients with **cachexia** (severe muscle wasting), tailored exercise plans can support independence and ease symptoms such as pain and breathlessness.

Crucially, **respiratory muscle training**-a subset of breathing exercises—has been found to significantly enhance breathing capacity, reduce dyspnoea, and improve emotional outlook in lung cancer patients.

The Challenge of Tailoring the Right Exercise

One of the key takeaways from the research studies is that there is no "one-size-fits-all" prescription. The **optimal type, intensity, and frequency** of exercise remains unclear and must be tailored to the individual's health status, cancer stage, and treatment plan.

A general guideline endorsed by the American College of Sports Medicine (ACSM) recommends:

- **50–300 minutes per week** of moderate-intensity cardiorespiratory activity (e.g., walking, cycling),
- **2–3 sessions per week** of resistance training,
- Inclusion of flexibility and balance exercises, especially for older or mobility-limited patients.

The concept of **progressive overload**-gradually increasing the exercise intensity and volume-was also emphasized to ensure continual improvement while minimizing risk.

Addressing Barriers to Exercise

Despite the proven benefits, the research also highlights that few lung cancer patients currently access exercise programs. Multiple barriers exist, including:

- Fatigue and side effects from treatment,
- Lack of access to exercise services, particularly in rural areas,
- Limited referrals from clinicians due to awareness gaps or resource constraints,
- Financial limitations, especially in healthcare systems where exercise programs are not reimbursed.

Stigma associated with lung cancer—often wrongly seen as a "self-inflicted" disease—can also lead to social withdrawal and reluctance to engage in supportive therapies like exercise.

To overcome these obstacles, the authors call for greater education among healthcare providers and patients about the role of exercise in managing lung cancer. They recommend embedding exercise specialists into cancer care teams and improving referral pathways.

A Multidisciplinary Approach is Key

Integrating exercise into the broader interdisciplinary care model is essential. The authors advocate for involving oncologists, exercise physiologists, dietitians, psychologists, and physiotherapists in developing personalized plans for each patient.

Special attention is warranted for patients with **bone metastases, comorbidities like COPD,** or those undergoing intensive therapies. For such individuals, the safety of exercise must be closely monitored. Supervised programs with proper risk assessments have been found to be both feasible and effective.

Conclusion

Exercise should no longer be considered an optional add-on in lung cancer care—it must be seen as an **essential component of treatment.** When designed thoughtfully and delivered by trained professionals, exercise offers patients the opportunity to reclaim control over their bodies, enhance their resilience, and improve their quality of life.

References:
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